

Titanic Dataset - EDA Report

1. Dataset Overview

The Titanic dataset consists of passenger data including demographics and survival status. The key columns include 'Survived', 'Pclass', 'Sex', 'Age', 'Fare', and 'Embarked'. The dataset has 891 rows and 12 columns.

2. Missing Values Handling

Missing values were found in the 'Age', 'Cabin', and 'Embarked' columns. 'Age' was filled with the median value. 'Embarked' was filled with the mode. 'Cabin' had too many missing values and was filled with 'Unknown'.

3. Univariate Analysis

Histograms and count plots were used to analyze the distribution of numerical and categorical features. 'Age' and 'Fare' are right-skewed. Most passengers were in the 3rd class, and most were male.

4. Bivariate Analysis

Boxplots showed that survivors tended to be younger and paid higher fares. Survival rates were higher among females. Heatmaps revealed moderate correlation between family size features (SibSp, Parch).

5. Handling Skewed Data

The 'Fare' feature was highly skewed. A log transformation ($\log(x+1)$) was applied, significantly reducing the skewness.

6. Multicollinearity Detection

Variance Inflation Factor (VIF) analysis showed no severe multicollinearity among numerical features like Age, Fare, SibSp, and Parch.

7. Key Findings

- Females had a higher survival rate than males.
- Passengers in 1st class had higher chances of survival.
- Fare and Age played a significant role in survival.
- Embarked port 'C' showed slightly higher survival rates.

8. Conclusion

The EDA provided valuable insights into survival factors on the Titanic. Visualizations, statistical methods, and feature engineering were effectively used to clean the data and extract meaningful patterns.